Connection-independent rescaling limit of the Finsler metric on jet bundles

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned journal source file sources/ahl-197-on-the-existence-of-logarithmic-and-orbifold-jet-2024/pinned-original-source.tex (On the existence of logarithmic and orbifold jet differentials); no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: Statement asserts the rescaling commutes with the C* action and the error factor is 1 plus or minus O(max_{2<=s<=k} (eps_s/eps_{s-1})^s); published printed number is Lemma 4.1. The only labelled reference inside the span is the lemma's own label (self-citation at the end of the proof, resolves internally); no cited works and no labelled equations inside the span.

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Editorial scope. the counted statement and its own proof environment only

Lemma 4.1. *Let us consider the global C^∞ bundle isomorphism $J^kV \to V^{\oplus k}$ associated with an arbitrary global C^∞ connection ∇ on $V \to X$, and let us introduce the rescaling transformation*

$$\rho_{\nabla,\varepsilon}(\xi_1,\xi_2,\ldots,\xi_k) = \big(\varepsilon_1^1\xi_1, \varepsilon_2^2\xi_2, \ldots, \varepsilon_k^k\xi_k\big) \quad \text{on fibers } J^kV_z,\ z\in X.$$

Such a rescaling commutes with the $\mathbb{C}^*$-action. Moreover, if p is a multiple of $\operatorname{lcm}(1,2,\ldots,k)$ *and the ratios $\varepsilon_s/\varepsilon_{s-1}$ tend to 0 for all $s=2,\ldots,k$, the rescaled Finsler metric $\Psi_{h,b,\varepsilon}\circ\rho^{-1}_{\nabla,\varepsilon}(\xi_1,\ldots,\xi_k)$ converges towards the limit*

$$\left(\sum_{1\leqslant s\leqslant k}\|\xi_s\|_h^{2b/s}\right)^{1/b}$$

on every compact subset of $V^{\oplus k} \setminus \{0\}$*, uniformly in* C^∞ *topology, and the limit is independent of the connection* ∇*. The error is measured by a multiplicative factor* $1 \pm O(\max_{2 \leqslant s \leqslant k} (\varepsilon_s/\varepsilon_{s-1})^s)$*.*

Proof. Let us pick another C^∞ connection $\widetilde{\nabla} = \nabla + \Gamma$ where $\Gamma \in C^\infty(U, T^*_X \otimes \mathrm{Hom}(V,V))$. Then $\widetilde{\nabla}^2 f = \nabla^2 f + \Gamma(f)(f') \cdot f'$, and inductively we get

$$\widetilde{\nabla}^s f = \nabla^s f + P_s\left(f\,;\, \nabla^1 f,\, \ldots,\, \nabla^{s-1} f\right)$$

where $P(z\,;\,\xi_1,\,\ldots,\,\xi_{s-1})$ is a polynomial with C^∞ coefficients in $z \in U$, which is of weighted homogeneous degree s in $(\xi_1,\,\ldots,\,\xi_{s-1})$. In other words, the corresponding isomorphisms $J^k V \simeq V^{\oplus k}$ correspond to each other by a $\mathbb{C}^*$-homogeneous transformation $(\xi_1,\,\ldots,\,\xi_k) \mapsto (\widetilde{\xi}_1,\,\ldots,\,\widetilde{\xi}_k)$ such that

$$\widetilde{\xi}_s = \xi_s + P_s\left(z\,;\,\xi_1, \ldots, \xi_{s-1}\right).$$

Let us introduce the corresponding rescaled components

$$(\xi_{1,\varepsilon}, \ldots, \xi_{k,\varepsilon}) = \left(\varepsilon_1^1 \xi_1,\ \ldots,\ \varepsilon_k^k \xi_k\right), \qquad (\widetilde{\xi}_{1,\varepsilon},\,\ldots,\,\widetilde{\xi}_{k,\varepsilon}) = \left(\varepsilon_1^1 \widetilde{\xi}_1,\,\ldots,\,\varepsilon_k^k \widetilde{\xi}_k\right).$$

Then

$$\begin{aligned}\widetilde{\xi}_{s,\varepsilon} &= \xi_{s,\varepsilon} + \varepsilon_s^s\, P_s\left(x\,;\,\varepsilon_1^{-1}\xi_{1,\varepsilon},\,\ldots,\,\varepsilon_{s-1}^{-(s-1)}\xi_{s-1,\varepsilon}\right)\\ &= \xi_{s,\varepsilon} + O(\varepsilon_s/\varepsilon_{s-1})^s\, O\left(\|\xi_{1,\varepsilon}\| + \cdots + \|\xi_{s-1,\varepsilon}\|^{1/(s-1)}\right)^s\end{aligned}$$

and it is easily seen, as a simple consequence of the mean value inequality $|\|x\|^\gamma - \|y\|^\gamma| \leqslant \gamma \sup_{z\,\in\,[x,y]} \|z\|^{\gamma-1}\|x-y\|$, that the "error term" in the difference $\|\widetilde{\xi}_{s,\varepsilon}\|^{2b/s} - \|\xi_{s,\varepsilon}\|^{2b/s}$ is bounded by

$$(\varepsilon_s/\varepsilon_{s-1})^s \left(\|\xi_{1,\varepsilon}\| + \cdots + \|\xi_{s-1,\varepsilon}\|^{1/(s-1)} + \|\xi_{s,\varepsilon}\|^{1/s}\right)^{2b}.$$

When b/s is an integer, similar bounds hold for all derivatives $D^\beta_{z,\xi}(\|\widetilde{\xi}_{s,\varepsilon}\|^{2b/s} - \|\xi_{s,\varepsilon}\|^{2b/s})$ and the Lemma 4.1 follows. □